

Final Exam

3/15

Sciences Lecture Hall 123

Monday, March 19 1:00pm

Topics 11.1 Linear Systems Theory

11.2.1 Compartment Models

11.3 Nonlinear Systems

11.4.4 A Mathematical Model for Neuron Activity

12.1 Counting

12.2 Probability

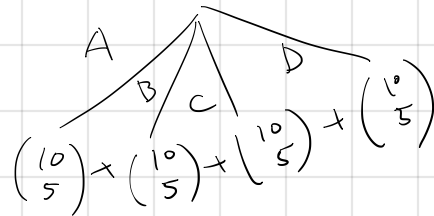
12.3.1 Conditional Probability

Homework 8 Problem 2

11.4.4,

a)

$$4 \cdot \binom{10}{5}$$



$$\Omega = \{HH, HT, TH, TT\}$$

$$P(\{HH\}) = \frac{1}{4}$$

$$P(\{HH\}^c) = P(\Omega) - P(\{HH\})$$

$$= 1 - \frac{1}{4} = \frac{3}{4}$$

$$b) \binom{40}{5} - 4 \cdot \binom{10}{5}$$

$$P(\Omega) = 1, P(\emptyset) = 0$$

$$P(\Omega) = 1$$

$$A \subseteq \Omega \quad 0 \leq P(A) \leq 1$$

$$P(A)$$

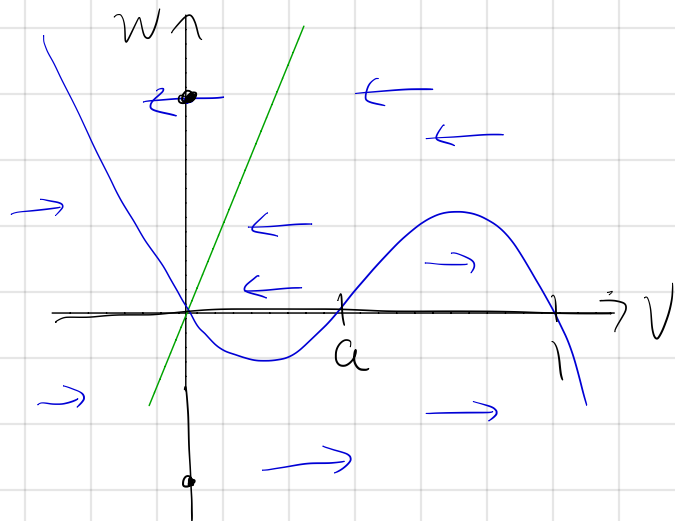
$$P(A \cup B) = P(A) + P(B)$$

$$A^c$$

$$\frac{dV}{dt} = -V(V-a)(V-1) - w \quad \underline{\quad} = 0 \quad w = -V(V-a)(V-1)$$

$$\frac{dw}{dt} = b(V - cw) \quad \underline{\quad}$$

$$w = \frac{1}{c} V$$



$$(w > 0, V = 0) \quad \frac{dV}{dt} = -w$$

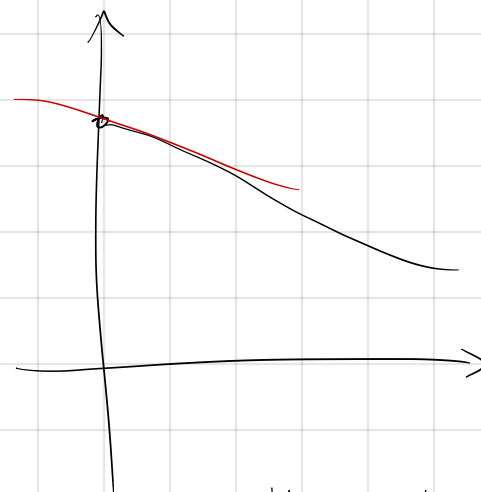
$$(w < 0, V = 0) \quad \frac{dV}{dt} = -w$$



action potential

$$\frac{dV}{dt}(0) > 0$$

$$1 > V(0) > a$$



no action potential

$$\frac{dV}{dt} \leq 0$$

$$0 < V(0) < a$$

What is the probability of exactly 2 pairs without replacement?

1 2 1 2

$$\frac{\binom{13}{2} \cdot \binom{5}{4} \cdot \binom{44}{1}}{\binom{52}{5}}$$

pairs

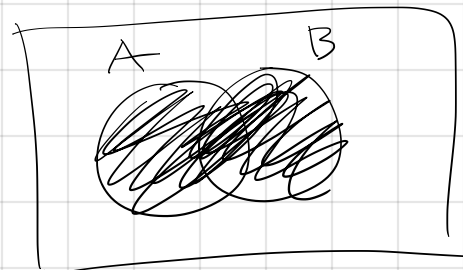
$$\binom{5}{2} \binom{3}{2}$$



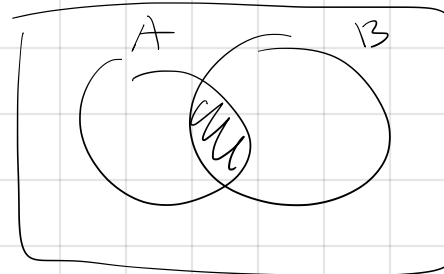
$$2. P(A|B) = \frac{P(A \cap B)}{P(B)}$$

$A \cup B$ - "or" XOR

$A \cap B$ - "and"



$A \cup B$



$A \cap B$

$$P(A|B) = \frac{P(A \cap B)}{P(B)} = \frac{\frac{1}{16}}{\frac{1}{4}} = \boxed{\frac{1}{4}}_{2^4}$$

$$P(B) = \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}$$

$$P(D) = \frac{1}{2} =$$

$$P(C) = \frac{1}{2}$$

$$\underline{2} \underline{2} \underline{1} \underline{2} = 2^3$$

$$\frac{2^3}{2^4} = \frac{1}{2}$$

$$P(D \cap C) = P(D)P(C) = \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}$$

$$P(A \cap B) = P(A) = \frac{1}{2^4} = \frac{1}{16}$$

$$B \subset A$$